

Semester 2 Final Exam — Bank-Draw Specification

Internal build document — not student-facing. Implements the finals design confirmed in `plan.md` §8 for PROBABILITY & STATISTICS HONORS (Semester 2, Week 36). Mirrors `semester-1-finals/finals-quiz-spec.md`.

1. Exam summary

Property	Value
Canvas item	Classic Quiz (graded), "Semester 2 Final Exam"
Total questions	50
Total points	50 (1 pt per question)
Source	Semester 2 lesson-quiz banks only — no new items
Draw method	One question group per module, random draw from that module's bank
Time limit	90 minutes (suggested; see §5)
Placement	Semester 2 Finals module, after the review lesson and review activity

Every question on this final comes from a lesson quiz the student has already taken. That is deliberate and is the **fairness rationale**: the lesson quizzes were announced all semester as the finals bank, so the final contains no unseen formats, no unseen difficulty, and no gotchas — only a comprehensive, proportionally weighted re-draw of material every student has already practiced under quiz conditions. Preparation reduces to honest review of one's own quiz history, which is exactly the study behavior the course wants to reward (and is what the review lesson's "how to prepare" plan teaches).

2. Question banks — naming convention

Create (or verify) six Canvas question banks, one per module, named exactly:

Bank name	Contents	Items
PSH S2 • Module 7 Bank	All lesson-quiz items, weeks 19–21 (Data & Data Displays)	45
PSH S2 • Module 8 Bank	All lesson-quiz items, weeks 22–24 (Descriptive Statistics)	45
PSH S2 • Module 9 Bank	All lesson-quiz items, weeks 25–26 (The Normal Distribution)	30
PSH S2 • Module 10 Bank	All lesson-quiz items, weeks 27–29 (Bivariate Data & Regression)	45
PSH S2 • Module 11 Bank	All lesson-quiz items, weeks 30–32 (Sampling & Study Design)	45
PSH S2 • Module 12 Bank	All lesson-quiz items, weeks 33–35 (Inference & Statistical Reasoning)	45

Bank total: 255 items (17 weeks × 3 quizzes × 5 questions). Source of truth for item text: each week's `lesson-quizzes.md` file under `module-07-...` through `module-12-...`. As lesson quizzes are built in Canvas, file every question into its module bank at creation time — do not defer the filing to finals week. (The quiz files name their bank as, for example, "Module 10 bank"; use the PSH S2 • Module N Bank names above in Canvas.)

Naming rules: prefix PSH S2 (course + semester), middle dot separator, Module N Bank (N = 7–12), mirroring Semester 1's PSH S1 • Module N Bank.

3. Question groups and draw counts

Build the final as **six question groups**, each linked to its module bank ("Link to Question Bank" inside the group). Draws are proportional to module size (3-week modules draw 9; the 2-week Module 9 draws 6; Module 12, a 3-week module, draws 8 so the exam totals exactly 50), per `plan.md` §8:

Group (order on exam)	Linked bank	Bank size	Draw	Pts/question	Group pts
Group 1	PSH S2 • Module 7 Bank	45	9	1	9
Group 2	PSH S2 • Module 8 Bank	45	9	1	9
Group 3	PSH S2 • Module 9 Bank	30	6	1	6
Group 4	PSH S2 • Module 10 Bank	45	9	1	9
Group 5	PSH S2 • Module 11 Bank	45	9	1	9
Group 6	PSH S2 • Module 12 Bank	45	8	1	8
Total		255	50		50

These counts match the "how to prepare" table in the Semester 2 review lesson (9 / 9 / 6 / 9 / 9 / 8); if the draw changes, change that table too. Because each group draws randomly from a bank 5–5.6× the draw size (Modules 7–11 at 5×, Module 12 at 5.6×), every student receives a different but proportionally comprehensive form, and item-order memorization between students is useless. Group order (M7 → M12) keeps the exam's arc parallel to the semester's; within each group Canvas randomizes which items appear.

4. Quiz settings

Setting	Value	Note
Quiz type	Graded quiz	
Shuffle answers	ON	Defeats choice-position memorization, and it matters here: in the audit in §9 the correct answer sits in position b for 86 items and c for 76 items, but only a for 51 and d for 42, so unshuffled forms would reward guessing b or c. Safe for this bank: no item uses "all/none of the above" or position-dependent choices (verified in §9).
Time limit	90 minutes	Suggested — see §5.
Allowed attempts	1	It is a final.
One question at a time	Recommended: ON	Reduces copy-paste of whole forms and keeps focus; pairs with the backtracking decision below.
Lock questions after answering (no backtracking)	OFF by default — teacher's choice	OFF is the student-friendlier setting (allows review before submit). A teacher may turn it ON for stricter integrity; either is acceptable, but decide before publishing and do not change between students.
Show correct answers	After due date (or after all submissions)	Prevents answer leakage while the window is open.
Access	No access code needed for the default asynchronous administration	

5. Timing guidance

Suggested window: 90 minutes for 50 questions (≈ 1 min 45 s per question). The items are 5-question lesson-quiz items — short by construction — so 90 minutes is deliberately generous: it accommodates computation with the Desmos Scientific Calculator (desmos.com/scientific) and, where a question needs a normal-curve area or an interval endpoint, a spreadsheet formula, plus honest re-reading. Students should have both open in other tabs, exactly as on every lesson quiz all semester.

6. Implementation steps (Canvas Classic Quizzes)

1. **Verify the banks.** Course → Quizzes → : → *Manage Question Banks*. Confirm the six PSH S2 • Module N Bank banks exist with the §2 item counts (45/45/30/45/45/45). If any lesson-quiz items were never filed into banks, move them in now

(open each lesson quiz → question → *move/copy to bank*).

2. **Create the quiz.** Quizzes → + Quiz → Classic Quiz. Title: Semester 2 Final Exam. Instructions: brief student-facing text (see §7). Quiz type: Graded Quiz. Assignment group: Exams/Finals per course grading setup.
3. **Apply settings** from §4 (shuffle answers ON, time limit 90, 1 attempt, one-question-at-a-time ON, backtracking per teacher's decision, correct answers shown after due date).
4. **Build the six groups.** Questions tab → *+ New Question Group*, six times, in module order. For each group: name it (Module 7 – draw 9, Module 9 – draw 6, Module 12 – draw 8 etc.), set *Pick* to the §3 draw count, set *points per question* to 1, then click **Link to Question Bank** and select the matching PSH S2 · Module N Bank. Do not hand-pick questions; the link + pick count does the random draw per student.
5. **Verify the total.** The quiz should display 50 questions / 50 points. If it shows more or fewer, a group's pick count or a stray loose question is the culprit.
6. **Preview.** Use *Preview* at least twice — confirm the draws differ between previews, math renders correctly (symbols such as $\hat{y}$, μ , σ , $\bar{x}$, $\hat{p}$ and exponents), and shuffled choices read sensibly on every item type.
7. **Publish** with the due date and the availability window for finals week. One quiz serves all students; the per-student randomization happens at attempt time.

7. Suggested student-facing instructions (paste into the quiz description)

Fifty questions, fifty points, one attempt, 90 minutes. Every question on this exam is drawn from the lesson quizzes you have already taken this semester — nothing here is new. Questions come from every module, in proportion to its size. Have your Desmos Scientific Calculator (desmos.com/scientific) and Excel open in other tabs, and pencil and paper at hand for summaries, tables, and interval work. Answer choices are shuffled; read each choice fully. When you finish, you are done with Probability and Statistics — well earned.

8. Maintenance notes

- **Do not edit bank items after any student has begun the final** — Classic Quizzes resolves bank draws at attempt time, and mid-window edits create inconsistent forms. Fix erroneous items only between terms (or regrade via Canvas's regrade tools if unavoidable).
- If a lesson-quiz item is later found defective and removed from a bank, keep the bank $\geq 2\times$ its draw count (all six banks currently sit at 5–5.6 $\times$; Modules 7–11 sit at exactly 5 $\times$ and Module 12 at 5.6 $\times$, so all are equally tight).
- Several Week 33–35 items reference simulated or "made-up" scenarios with fixed numbers (shuffle counts, interval endpoints); none depends on a widget being re-run, so no item changes between sittings.
- Item defects reported by students or teachers should be recorded against the week's `lesson-quizzes.md` file (the source of truth), then fixed in Canvas between terms.

9. Standalone audit of the 255 bank items (build check, F.3)

Run on 2026-10-07 against every `lesson-quizzes.md` in `module-07-...` to `module-12-...` (17 files, weeks 19–35) with a script (`audit.py`, kept in the build scratchpad, not in the repo) plus manual reading of every flagged item.

Check	Result
Files / quizzes / items	17 weeks $\times$ 3 quizzes $\times$ 5 items = 255 ; per module 45 / 45 / 30 / 45 / 45 / 45 (matches §2)
Every item has exactly four choices (a–d) and exactly one flagged correct answer	Pass, 255 of 255
Item type	All multiple choice; no open response
Correct-answer positions	a 51, b 86, c 76, d 42 (uneven; shuffle answers must be ON — §4)
"All / none of the above", "both a and b", references to "the lesson", "the widget", "the figure", "Week N", "Lesson N", "question N", "the previous question"	None found (the script's hits were all ordinary uses of "above/below" as comparisons, e.g. "scored above 90", "marks above 2", "the line above"; the one "the lesson" hit is a study lesson inside a made-up experiment)

References to unseen data ("the table shown", "this graph", "these data" without the data)	None found ; the few items saying "these data" give the data in the stem (checked Weeks 28, 29 and 24 by reading)
Scenarios	Made up / constructed / simulated and labeled as such in the stems; no real data, people or studies presented as fact
Worked solutions inside question text	Found and fixed during the C.2 export step. 28 items in weeks 19–23, 25 and 26 carried a line <code>*(Work: ...)*</code> between the stem and the choices that showed the worked solution (answer included). The first audit missed it. Each line is now an internal HTML comment (<code><!-- Worked solution (internal, not for students): ... --></code>) in the <code>lesson-quizzes.md</code> files, so it is neither pasted into Canvas by hand nor exported to QTI. If you entered any of these quizzes into Canvas before this fix, delete the line from the question text.
Internal alignment and answer-key notes	Kept in markdown comments and header lines of the <code>.md</code> files only; never paste them into Canvas

Format note for the builder: Week 29's `lesson-quizzes.md` uses single-# quiz headings (`# Quiz 29.2`, `# Quiz 29.3`) where every other week uses `##`; this affects only the markdown, not the items. Items in several weeks use markdown emphasis (`*x*`) for variables; convert to Canvas math/italics when entering them.

Not audited by this build step: the correctness of each item's answer was verified when each week was built (every numeric answer recomputed by script at build time, key values re-checked by the lead); this step checked structure and standalone wording, not a second full recomputation of all 255 items.

Semester 2 Final Exam: the fixed 50-question form in the Course Kit

Internal build document, not student-facing. The Course Kit cartridge cannot carry bank draws, so the kit ships one fixed form: one question from each lesson quiz (Modules 7-11) and eight of Module 12's nine lesson quizzes, $9/9/6/9/9/8 = 50$. Every question is an existing lesson-quiz item. Selection is seeded (seed 2026100901) by `_build/make_kit_spec_s2.py`, so the form can be rebuilt identically.

#	Module	Quiz item	Question (first words)
1	7	19.1 Q2	Which of these variables is categorical, even though its values are wr
2	7	19.2 Q2	A frequency table has categories with frequencies 12, 9, 6, and 3. Wha
3	7	19.3 Q2	You want to compare the number of students in each of 12 school clubs.
4	7	20.1 Q2	Which dataset is best displayed with a line plot?
5	7	20.2 Q2	A histogram uses bins in which each bin includes its left endpoint but
6	7	20.3 Q3	Which statement describes a weakness of a box plot?
7	7	21.1 Q5	In a (made-up) town, monthly ice-cream sales and monthly swimming-pool
8	7	21.2 Q1	A bar chart shows two bars, 52 and 55, on a vertical axis that starts
9	7	21.3 Q3	After inserting a column chart in Excel, which setting should you chec
10	8	22.1 Q3	The data set 20, 22, 25, 26, 27 has mean 24 and median 25. If the larg
11	8	22.2 Q1	Dana's score on a 50-point quiz is at the 80th percentile of her class
12	8	22.3 Q3	For the data set 3, 5, 6, 8, 9, 11, 12, 40, use the median-of-each-hal
13	8	23.1 Q2	When computing a standard deviation, why are the deviations from the m
14	8	23.2 Q3	Twelve made-up scores are 10, 11, 12, 12, 13, 14, 86, 87, 88, 88, 89,
15	8	23.3 Q2	An Excel summary of the commute times, in minutes, of 30 riders (a sam
16	8	24.1 Q4	A student surveys 12 students from School X and 12 students from Schoo
17	8	24.2 Q4	In a made-up list of 11 commute times in minutes, ten values lie betwe

18	8	24.3 Q4	A bar chart shows two group means, 52 and 55, with the vertical axis s
19	9	25.1 Q2	A machine fills boxes so that the weights are approximately normal wit
20	9	25.2 Q5	Which of the following data sets is the poorest candidate for using th
21	9	25.3 Q1	A machine fills jars so that the fill volumes are approximately normal
22	9	26.1 Q2	A large dataset (1,000 values) has a mean of 50 and a standard deviati
23	9	26.2 Q2	A common rule of thumb says a normal curve can reasonably approximate
24	9	26.3 Q1	The battery lifetimes of a made-up device are approximately normally d
25	10	27.1 Q4	Nine points lie close to a rising straight line, and for those nine po
26	10	27.2 Q3	A straight line passes through the points (2, 24) and (10, 8). Which e
27	10	27.3 Q2	A fitted line is $\hat{y} = 12.5 + 3.2x$, based on data with x from 1 to 10. W
28	10	28.1 Q2	A least-squares line is fitted to 8 constructed points. In order of x
29	10	28.2 Q4	A spreadsheet has the x values in cells A2:A11 and the y values in cel
30	10	28.3 Q3	In a made-up set of ten towns, the correlation between ice cream sales
31	10	29.1 Q5	A student wants to fit $y = a \cdot bx$ by taking $\ln$ of y, but one of the cons
32	10	29.2 Q4	Which situation is best summarized with a two-way frequency table rath
33	10	29.3 Q1	A made-up survey of 400 club members asked two questions. Of the 150 m
34	11	30.1 Q5	A club president asks the five friends sitting nearest her whether the
35	11	30.2 Q5	A student uses =RANDBETWEEN to pick her sample and computes a sample m
36	11	30.3 Q3	Which of the following is a non-probability sampling method?
37	11	31.1 Q3	The owner of a made-up website invites visitors to vote on whether a n
38	11	31.2 Q3	A made-up call-in poll of a town reports 70% in favor of a proposal, w
39	11	31.3 Q3	A made-up game claims its coin is fair. It is flipped 50 times and lan
40	11	32.1 Q2	A made-up researcher selects 300 students at random from a school of 1
41	11	32.2 Q5	In a made-up experiment, 60 subjects are randomly assigned to a new ex
42	11	32.3 Q1	A made-up researcher wants to know whether a new study routine causes
43	12	33.1 Q5	In a made-up town, 30% of residents own a bicycle. For random samples
44	12	33.2 Q2	A made-up poll of 250 randomly chosen people has a margin of error of
45	12	33.3 Q4	A made-up random sample of 64 plants has a mean height of 72 cm, and t
46	12	34.1 Q5	A researcher takes a random sample of 50 students who walk to school a
47	12	34.2 Q1	In a made-up randomized experiment, 40 seeds were assigned to each of
48	12	35.1 Q4	A test of a fair-coin claim uses a rule built to keep the false-alarm
49	12	35.2 Q3	A made-up headline says, "Program doubles reading time." The report's
50	12	35.3 Q4	A made-up report measures 40 children aged 6 to 12 and fits the line h

Teachers who want a different form per student can build the six question banks and the six question groups described in the Final Exam Build Spec instead; the 255 questions are in the lesson quizzes.